# Joshi Sayali Shripad

cs24mtech12003@iith.ac.in | sayali21joshi@gmail.com | +91 6305079024 | LinkedIn | GitHub

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.68 |
| St. Anthony's Junior College | Class XII (TSBIE) | 2020 | 97.80% |
| St. Peter's High School | Class X (SSC) | 2018 | 10.00% |

## PUBLICATIONS

Peddi Manognya, Lokendra Mandloi, Joshi Sayali Shripad, Sandipan Dandapat. "Values Are Not Enough: Situational Modifiers Shape Moral Decisions in LLMs and Humans." Findings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Budapest, Hungary. Accepted.

## RESEARCH

**zk-CCI: A Trustless, Verifiable Multi-Gateway Framework for Cross-Chain Interoperability |** *Blockchain, Zero-Knowledge Proofs, Distributed Systems*

- Developed a trustless multi-gateway framework using zk-SNARKs for verifiable cross-chain message validation.
- Implemented and evaluated a proof-of-concept on Ethereum and Hyperledger Fabric for EHR consent management.
- Enabled destination blockchains to cryptographically verify gateway validation without relying on trusted relays.
- Conducted the research in collaboration with DENSO Corporation, Japan

## PROJECTS

**VISTA: Situational Modifiers and Moral Decision-Making in LLMs | Github**

- Conducted a study on impact of situational context on moral decision-making in LLMs under fixed human value profiles.
- Evaluated Gemma 4, Qwen 2.5, and LLaMA 3.1 using a novel decision-flip metric and human comparison study.
- Co-authored a paper Accepted in EMNLP 2026

**MV-cBBDM: SAR-to-Optical Image Translation | Github**

- Reproduced cBBDM for VHR SAR-to-optical image translation and benchmarked performance on the MSAW dataset.
- Proposed a multi-view diffusion framework using cross-attention to fuse multiple SAR observations for improved reconstruction.
- Implemented VQ-VAE and Brownian Bridge diffusion, evaluated with LPIPS, FID, CW-SSIM, FSIMc, and SAM.

**Privacy-Preserving Petition System | Github**

- Developed a blockchain-based, anonymous yet verifiable petition platform using threshold credentials and zk proofs.
- Implemented Pedersen commitments, credential randomization, ECDSA verification, and nullifiers for double-sign prevention.
- Designed Solidity contracts and off-chain verification workflows with selective disclosure and zk-SNARK eligibility verification.

**Knowledge Graph Optimization for SDN-based Digital Twin Networks | Github**

- Designed an incremental Knowledge Graph framework for Digital Twin Networks, avoiding full graph reconstruction.

- Developed schema-driven node indexing and parallel graph-query execution to accelerate topology and flow-rule analysis.

- Achieved ~99% lower KG build time, ~91% smaller graph size, and ~98% faster flow-rule reachability queries.

**Miniature Bitcoin | Github**

- Built a simplified Bitcoin implementation with wallets, UTXO transactions, Proof-of-Work, and block mining.

- Implemented P2P transaction propagation, mempool management, block validation, Merkle trees, and digital signatures.

- Designed end-to-end transaction processing from wallet creation and broadcasting to mining and blockchain updates.

## SKILLS

**Programming Languages:** C/C++, Python, SQL, Solidity

**Machine Learning & AI:** PyTorch, Deep Learning, Machine Learning

**Blockchain & Cryptography:** Ethereum, Smart Contracts, Hyperledger Fabric, Zero-Knowledge Proofs (zk-SNARKs)

**Networking & Systems:** Wireshark, TCPdump, SDN, Knowledge Graphs (Neo4j)

**Tools & Technologies:** Git, Linux, Truffle, Ganache

## RELEVANT COURSES

Advanced Computer Networks | Advanced Data Structures and Algorithms | Blockchain Technologies | Foundations of Machine

Learning | Fraud Analytics Using Predictive and Social Network Techniques | Linear Optimization | Network Engineering |

Representation Learning | Topics in Deep Learning

## POSITIONS OF RESPONSIBILITY

**Volunteer, 1st Japan -India Forum**

Assisted in organizing academic and industry interactions between Japanese and Indian delegates

**Volunteer, ACM ARCS 2026 and ACM India Annual Event 2026**

Supported poster and lightning-event coordination, participant management, and logistics.